AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 12 · WORKBOOK

Speech and Audio Systems

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected speech-evidence record.

Explain stages, trace a conflict, audit a limit and design a revealing check.

Keep one habit: Retain reference/transcript/parsed input/result/spoken output, missing evidence and scope.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Visual Composition. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Prompt Evaluation. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Examples and Rubrics. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Visual Composition

Prompt Evaluation

Examples and Rubrics

W1 · Explain the stage mechanism

> Keep source modality, transcript and later claims separate.

Your evidence record

- Unresolved W2 case: transcriptfifteen boxes,inventory50,originalaudio unavailable,no declared authority. Preserve conflict;certainty unsupported;firsttechnicalfailure unknown. Separate pictured calculation has supplied reference13+7,transcript30+7,parsed30+7,calculator37,spoken37. Recognition is first mismatch;arithmetic is correct for its received input. Corrected transcript under the same later rules gives20.

- Reference for this exercise: open this week's student lesson, "Speech and Audio Systems", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Trace an unresolved source conflict

> Do not invent replay or an unstated authority policy.

Your evidence record

- Unresolved W2 case: transcriptfifteen boxes,inventory50,originalaudio unavailable,no declared authority. Preserve conflict;certainty unsupported;firsttechnicalfailure unknown. Separate pictured calculation has supplied reference13+7,transcript30+7,parsed30+7,calculator37,spoken37. Recognition is first mismatch;arithmetic is correct for its received input. Corrected transcript under the same later rules gives20.

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- Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

  Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Audit a stage-level conclusion

> A correct or authoritative stage does not validate every source or downstream claim.

Your evidence record

- Unresolved W2 case: transcriptfifteen boxes,inventory50,originalaudio unavailable,no declared authority. Preserve conflict;certainty unsupported;firsttechnicalfailure unknown. Separate pictured calculation has supplied reference13+7,transcript30+7,parsed30+7,calculator37,spoken37. Recognition is first mismatch;arithmetic is correct for its received input. Corrected transcript under the same later rules gives20.

- A report omits this qualification: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Design a revealing stage-specific check

> Preserve fixed conditions and the evidence needed to locate a mismatch.

Your evidence record

- Unresolved W2 case: transcriptfifteen boxes,inventory50,originalaudio unavailable,no declared authority. Preserve conflict;certainty unsupported;firsttechnicalfailure unknown. Separate pictured calculation has supplied reference13+7,transcript30+7,parsed30+7,calculator37,spoken37. Recognition is first mismatch;arithmetic is correct for its received input. Corrected transcript under the same later rules gives20.

- Reference for this exercise: open this week's student lesson, "Speech and Audio Systems", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Diagnose a new calculation trace independently

> Keep one first attempt before hints or feedback.

Read and explain independently

The independent task supplies the known phrase and transcript as classroom facts, not actual audio. Treat the parsed operands as90and6 with subtraction, as stated by the calculator trace. Check its arithmetic against that received input and separately recover the intended reference expression. Do not infer real recogniser accuracy or claim a replay was performed.

Known audio says “nineteen minus six”. **Transcription** says “ninety minus six”; the calculator returns84. Locate the first error and propose a check.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week’s change to your existing project record.

Your evidence record

Show transcripts and require confirmation before saving memory or using tools.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Trace the worked case for Speech pipelines and multimodal **evidence**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Speech pipelines and multimodal **evidence**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Audio says 'sixteen divided by four'; transcript says 'sixty divided by four'; calculator returns 15. Which stage first fails? What would a useful confirmation show?

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A speech tool hears fifteen as fifty, then calculates correctly using fifty.

Where did the error enter?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A recording says fifteen, but the transcript says fifty. The summary repeats fifty. Where did the error enter, and what should be checked first?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 12 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Audio says 'sixteen divided by four'; transcript says 'sixty divided by four'; calculator returns 15. Which stage first fails? What would a useful confirmation show?

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Additional practice · Resolve a bounded authority task

REFERENCE · SECTION 15

Additional practice · Resolve a bounded authority task

> The retained roomE/F task differs from the selected known-phrase calculation assessment.

Earlier task and saved response

- Transcript says room E; a current written approval says room F. The task requires current approved room. Written approval is declared authoritative; trace the response and the **transcription** limit.

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

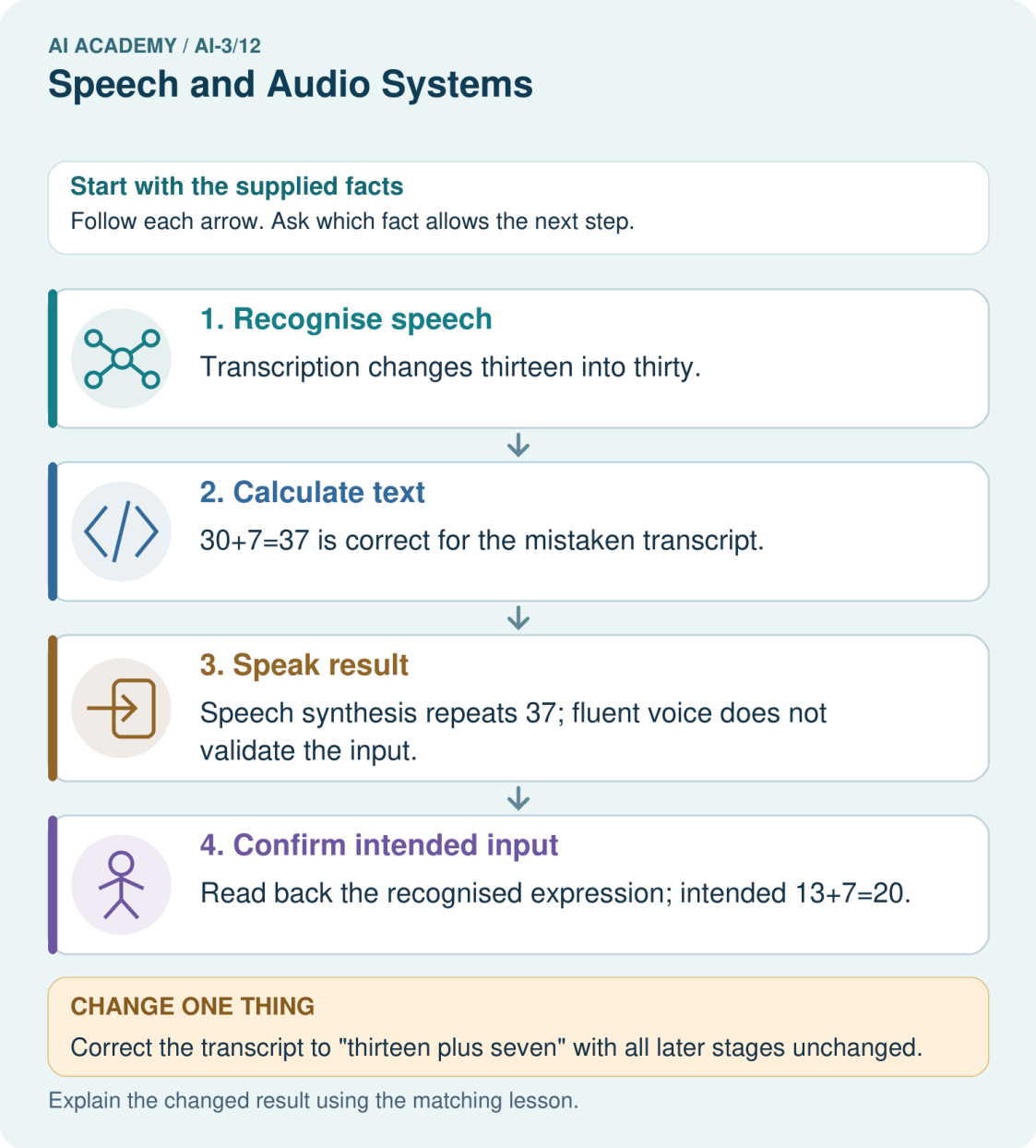
> Keep the explanation and the support you used.

Use this when helpful

A fictional learner dictates a calculation to a voice assistant.

Known audio phrase: “thirteen plus seven”. Transcript: “thirty plus seven”. Calculator returns 37.

Synthetic voice says “thirty-seven”. The example supplies ground truth; no real audio was analysed.



Follow the labelled picture with the matching explanation. Open the vector diagram

Which stage was initially wrong, and which stage was correctly operating on a wrong **input**?

> **My picture-based explanation and changed-case prediction**

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Show Study Buddy recognised questions before calculating or retrieving important answers, especially numbers and dates.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Verified Study Buddy

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Show Study Buddy recognised questions before calculating or retrieving important answers, especially numbers and dates.

Evidence to keep	My record
This week's input and deciding step	

Evidence to keep	My record
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional learner dictates a calculation to a voice assistant.
- Which labelled step matches this detail? Read back the recognised expression; intended 13+7=20.
- 1. Confirm intended **input**

- 2. Recognise speech
- 3. Calculate text
- 4. Speak result
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Confirm intended **input** Read back the recognised expression; intended 13+7=20. The first error occurs before arithmetic.

Blaming the calculator would target a stage that correctly **process**ed its supplied text. Read-back or on-screen confirmation lets the user catch a number misunderstanding before action.

Retain the original audio and transcript, where permitted, to separate recognition defects from later **process**ing errors.

Changed-case question

- Changed-case practice: Correct the transcript to “thirteen plus seven” with all later stages unchanged. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.

- 3. The calculator and spoken result should become 20 under the declared pipeline.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The calculator and spoken result should become 20 under the declared pipeline. 13+7=20; the changed **input** removes the earlier recognition error.

Repair and save your work

Review Visual Composition (ai-3/11). Goal: Check object counts, spatial relations and arrow meaning separately. Short practice: Which checks pass after that single change, and what still needs repair?. Return to this week and explain what you changed.

Review Prompt Evaluation (ai-3/9). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: Why is B's selection defensible here without calling it universally superior?.

Return to this week and explain what you changed.

Review Examples and Rubrics (ai-3/6). Goal: Distinguish a format example from approved factual evidence. Short practice: Explain why neither original nor changed draft is a full pass, and provide a complete repaired card.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

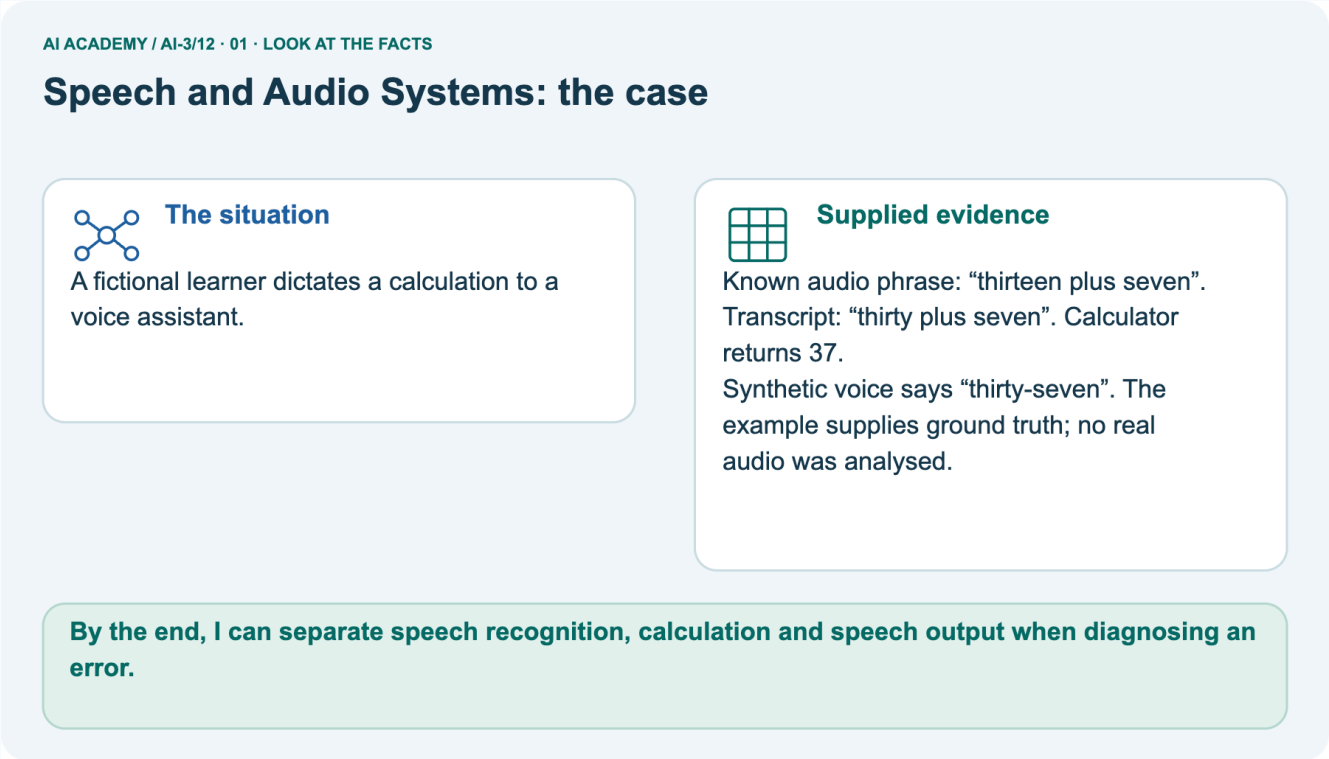
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Speech and Audio Systems: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

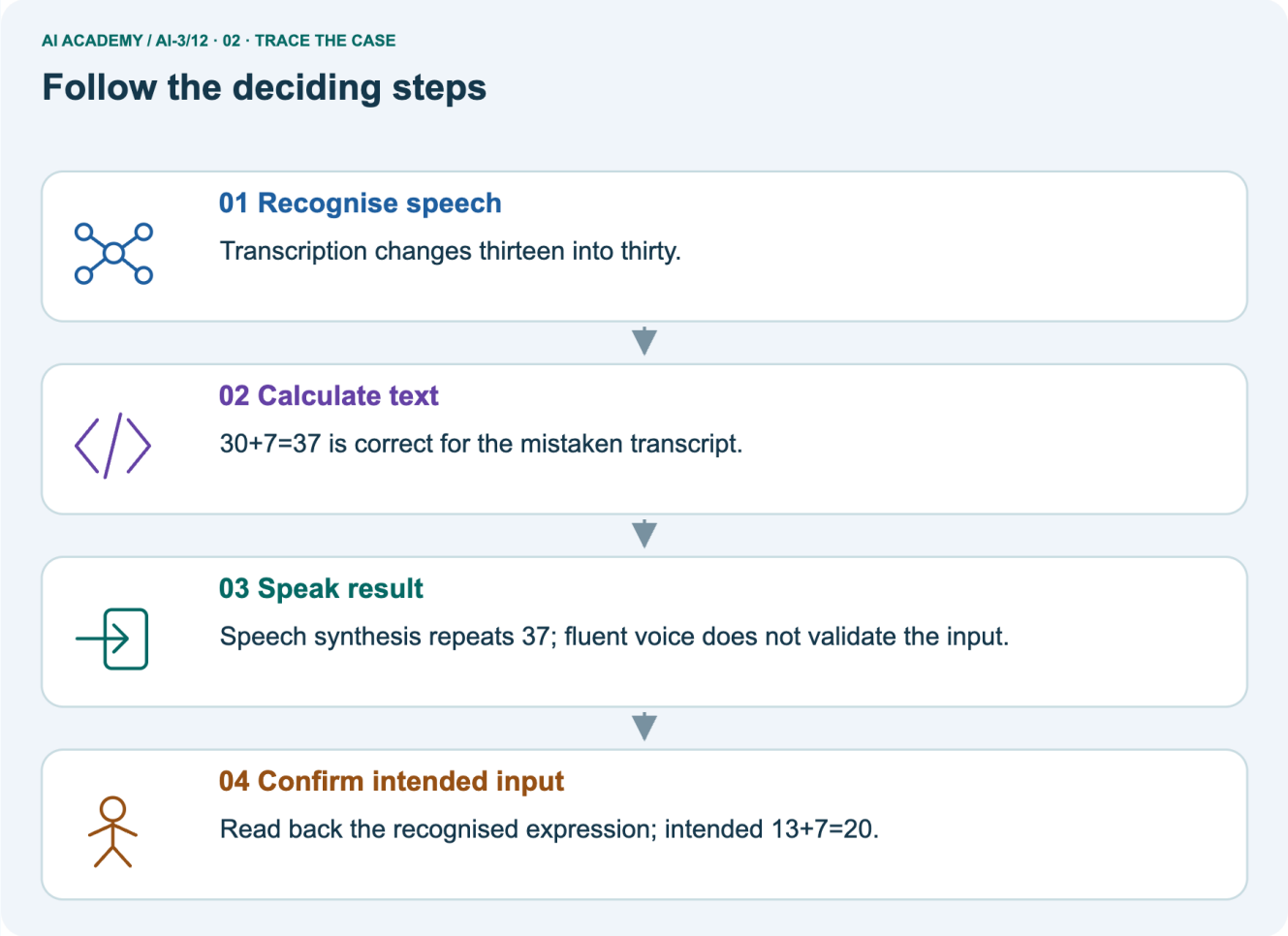
A fictional learner dictates a calculation to a voice assistant. Supplied facts: Known audio phrase: “thirteen plus seven”. Transcript: “thirty plus seven”. Calculator returns 37. Synthetic voice says “thirty-seven”.

The example supplies ground truth; no real audio was analysed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



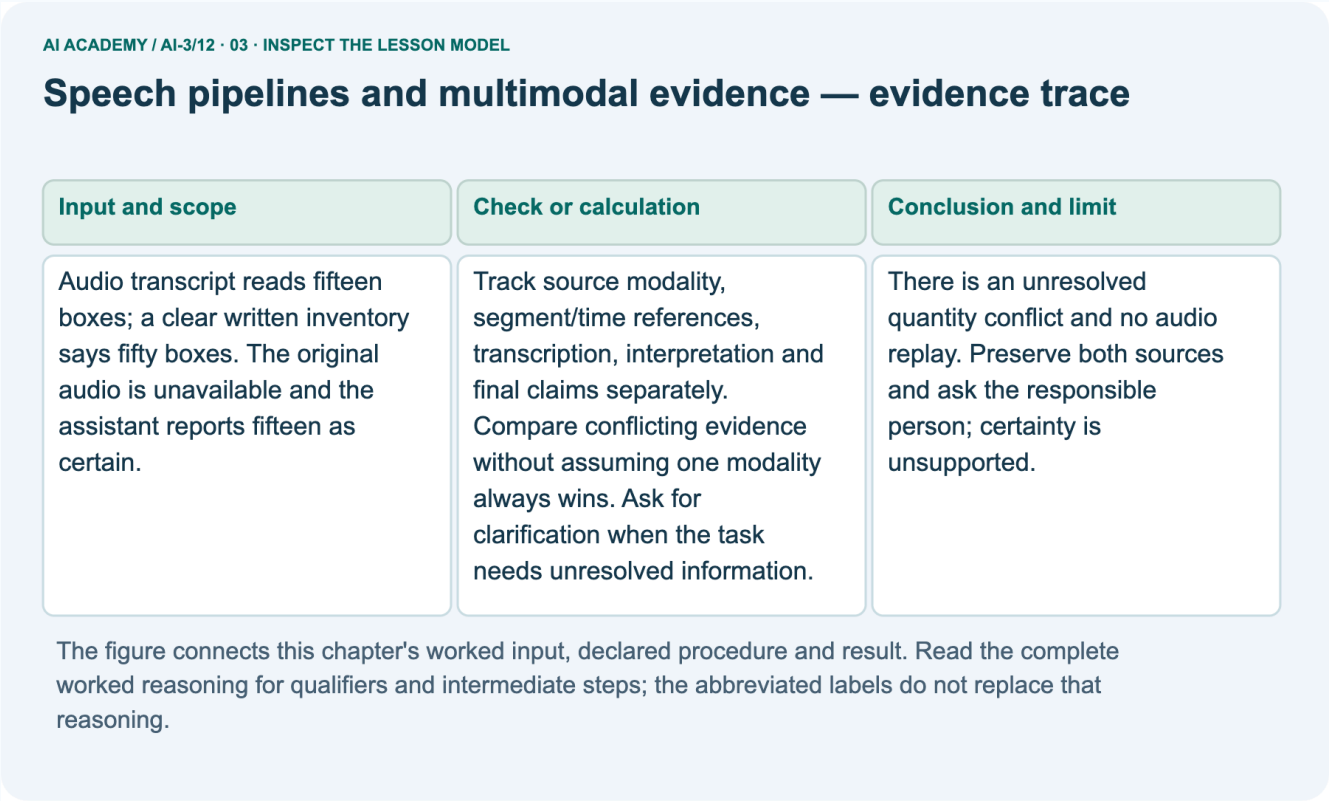
Follow the labelled picture with the matching explanation. Open the vector diagram

Recognise speech: **Transcription** changes thirteen into thirty. → Calculate text: 30+7=37 is correct for the mistaken transcript. → Speak result: Speech **synthesis** repeats 37; fluent voice does not validate the **input**. → Confirm intended **input**: Read back the recognised expression; intended 13+7=20.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Speech pipelines and multimodal evidence — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Speech pipelines and multimodal **evidence** — **evidence** trace. Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes.

The original audio is unavailable and the assistant reports fifteen as certain. | Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately.

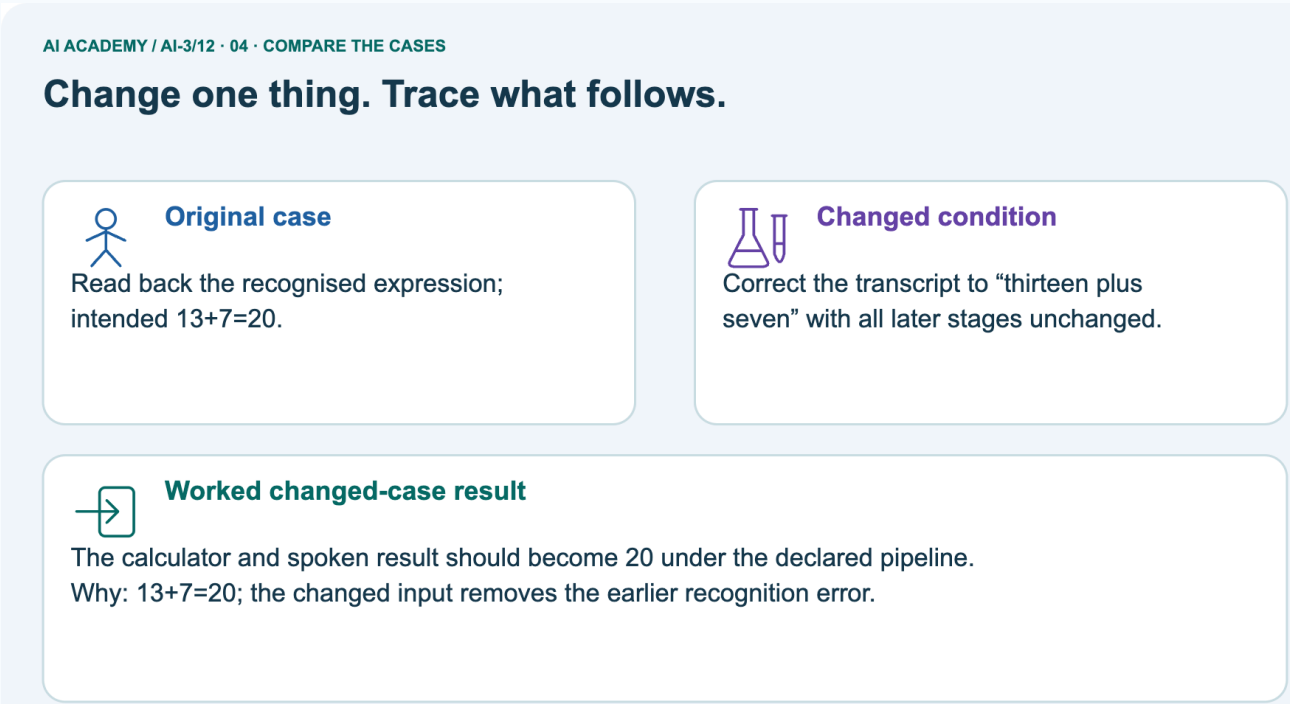
Compare conflicting **evidence** without assuming one **modality** always wins. Ask for clarification when the task needs unresolved information. | There is an unresolved quantity conflict and no audio replay.

Preserve both sources and ask the responsible person; certainty is unsupported.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

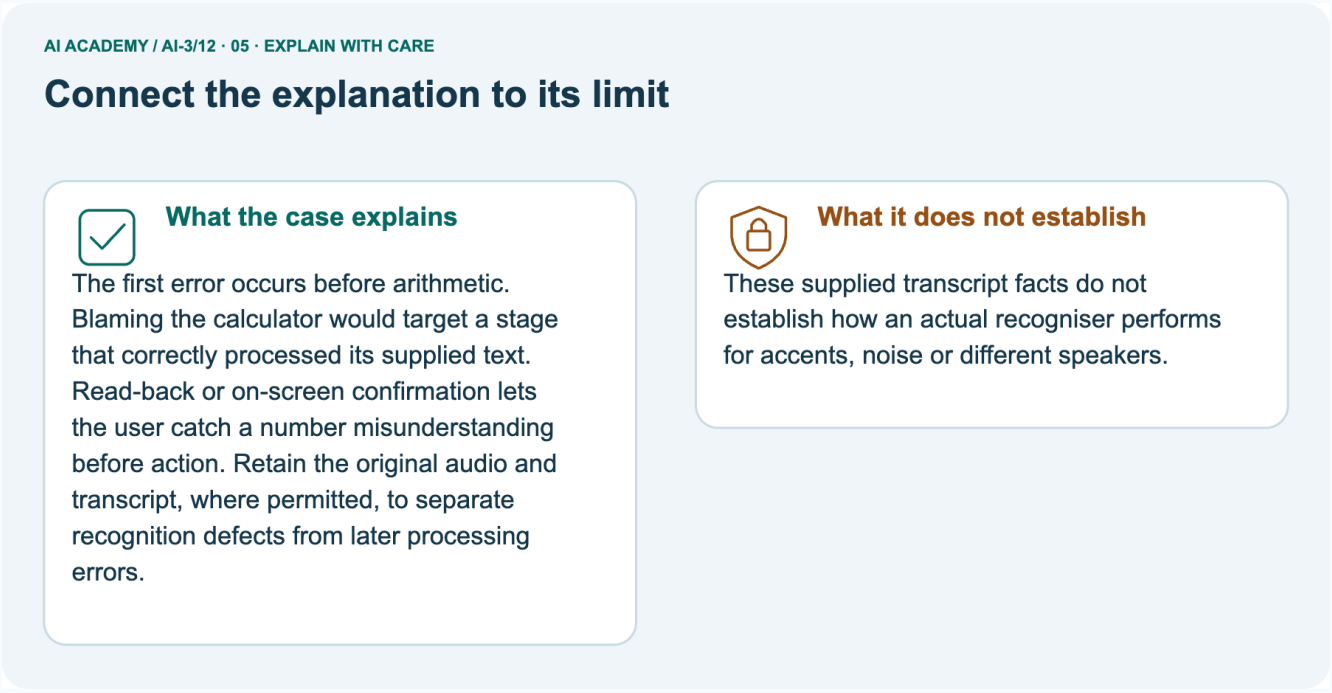
Original case: Read back the recognised expression; intended 13+7=20. Changed condition: Correct the transcript to “thirteen plus seven” with all later stages unchanged.

Expected result: The calculator and spoken result should become 20 under the declared pipeline. Reason: 13+7=20; the changed **input** removes the earlier recognition error.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The first error occurs before arithmetic. Blaming the calculator would target a stage that correctly **process**ed its supplied text. Read-back or on-screen confirmation lets the user catch a number misunderstanding before action.

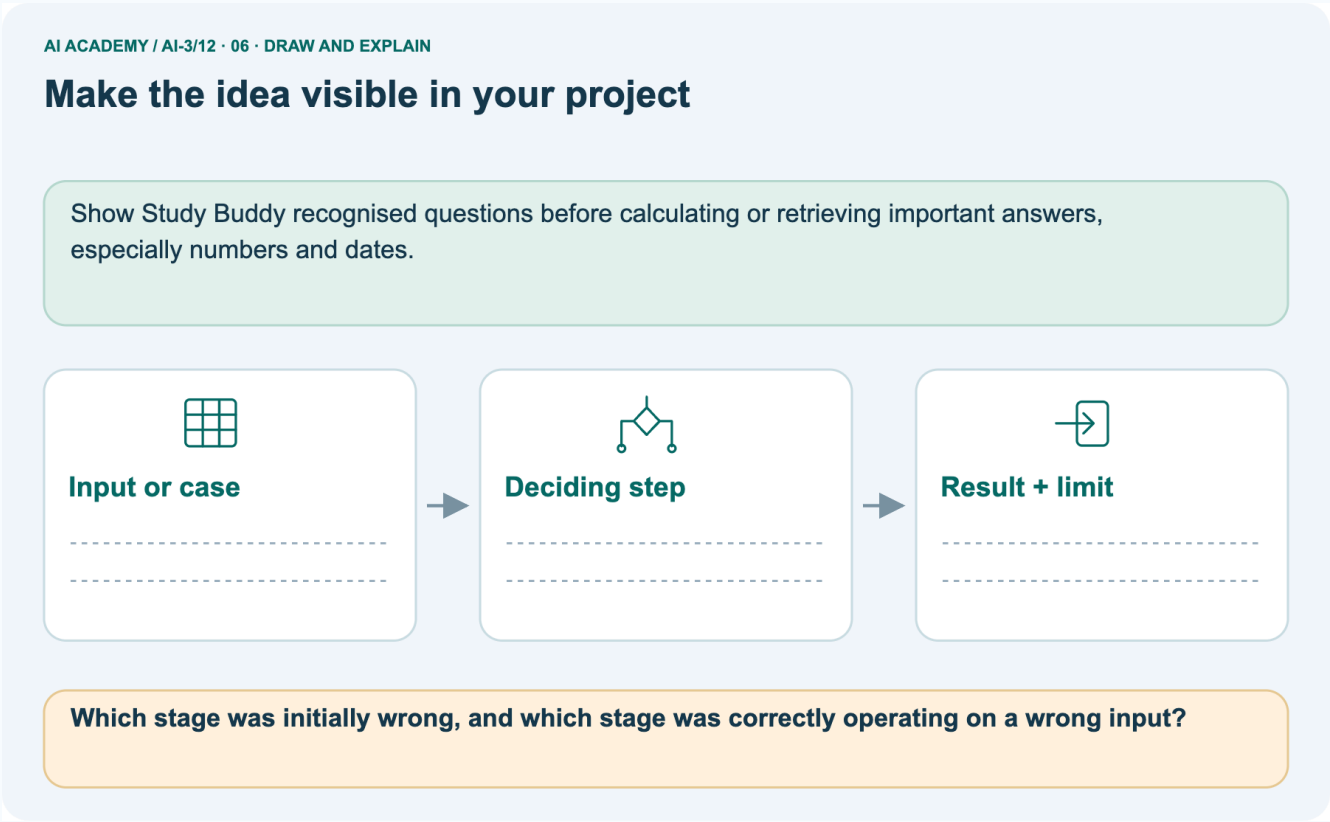
Retain the original audio and transcript, where permitted, to separate recognition defects from later **process**ing errors.

Limit: These supplied transcript facts do not establish how an actual recogniser performs for accents, noise or different speakers.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Show Study Buddy recognised questions before calculating or retrieving important answers, especially numbers and dates. Draw your **input**, deciding step, result and limit.

Supported question: Which stage was initially wrong, and which stage was correctly operating on a wrong **input**?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Speech and Audio Systems: the case

02. Follow the deciding steps

03. Speech pipelines and multimodal evidence — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.